Developing Coumarin-Imidazole Photocages for RNA

Deven Shah[1], Eric T. Kool[1]

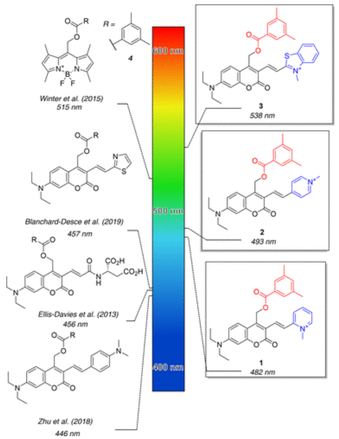
devens@stanford.edu

[1]Stanford University Department of Chemistry

Motivation

- Photoremovable protecting groups (PPGs), or photocages, temporarily block molecular activity. Light at a suitable wavelength restores activity. [1]
- UV-dependent RNA photocages face limits from phototoxicity and shallow optical penetration. [2,3]
- Longer-wavelength scaffolds could enable light-controlled RNA chemistry with less phototoxicity and greater tissue penetration.

Higher-Wavelength Scaffolds



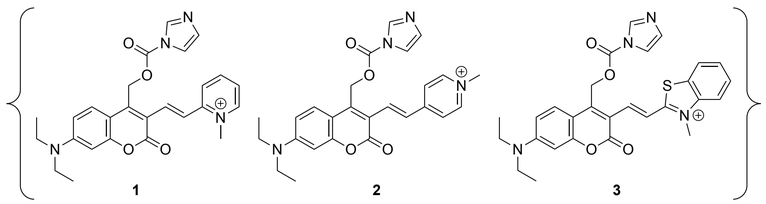
Visible-light scaffold design

- Coumarinylmethyl derivatives provide a tunable scaffold for shifting absorption toward longer wavelengths. [4]
- The design extends conjugation at the coumarin 3-position using heteroaromatic groups.
- Two-photon excitation and upconversion offer alternative approaches, but face efficiency limitations. [2,3]

Literature examples of photocage scaffolds and reported wavelengths. [4]

Goal: Coumarin-Imidazole RNA PPGs

Develop three coumarin-imidazole candidates for RNA photocaging by combining conjugation-extended chromophores with an imidazole activation group. [5]

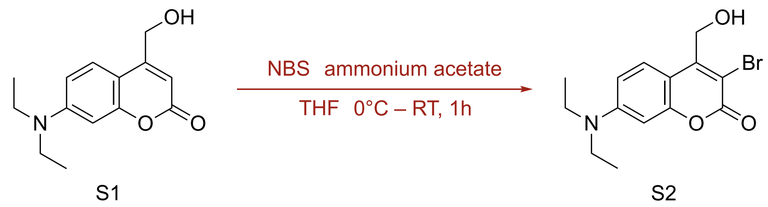


Target structures 1-3: 2-pyridyl, 4-pyridyl, and benzothiazolyl derivatives.

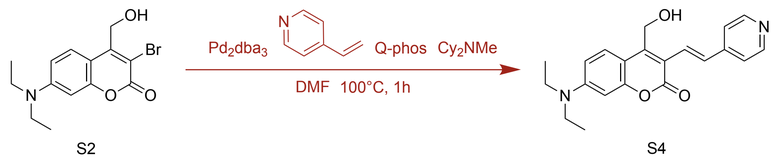
Synthetic Strategy

Representative route to the 4-pyridyl candidate (2). Related branches in the synthetic design lead to targets 1 and 3.

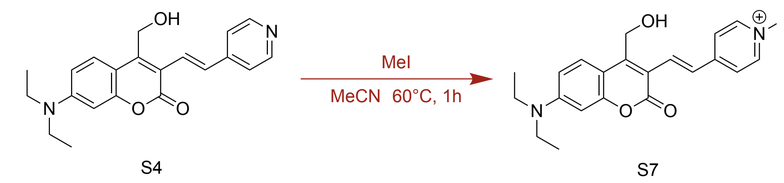
1. Bromination at C3



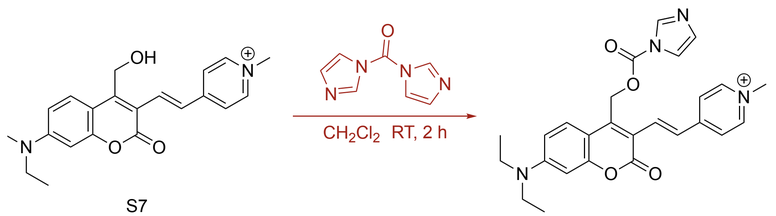
2. Heck coupling with 4-vinylpyridine



3. Pyridine N-methylation



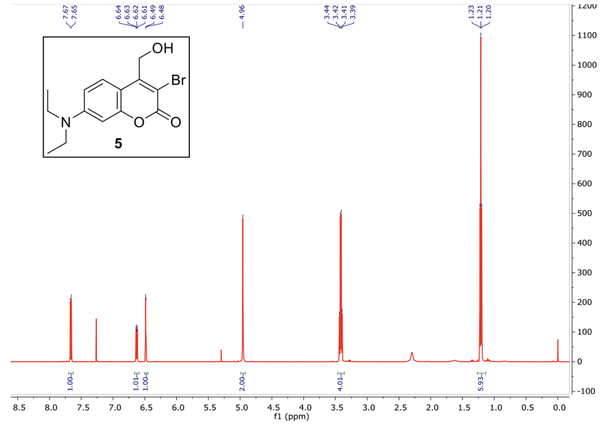
4. Imidazole-carbonate formation



Synthetic Progress

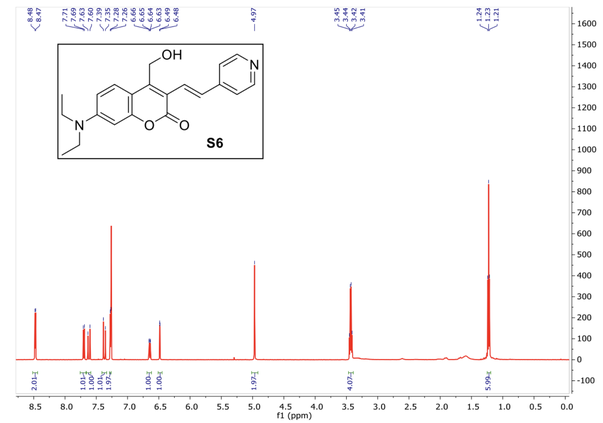
NMR spectra document the C3-brominated intermediate and the 4-pyridyl Heck product. The spectra below retain their original labels.

C3-brominated intermediate



Route S2; spectrum label 5. Chemical shift in ppm.

4-Pyridyl Heck product



Route S4; spectrum label S6. Chemical shift in ppm.

Next Steps

Complete and characterize the candidate PPGs, then measure photolysis wavelength dependence and test RNA caging and light-triggered release.

References

[1] J. Contr. Release 2019, 298, 154-156. [2] Angew. Chem. Int. Ed. 2013, 52, 4526-4537.
[3] Chem. Soc. Rev. 2017, 46, 1025-1039. [4] Org. Lett. 2019, 21, 9410-9414.
[5] J. Am. Chem. Soc. 2018, 140, 3491-3495.